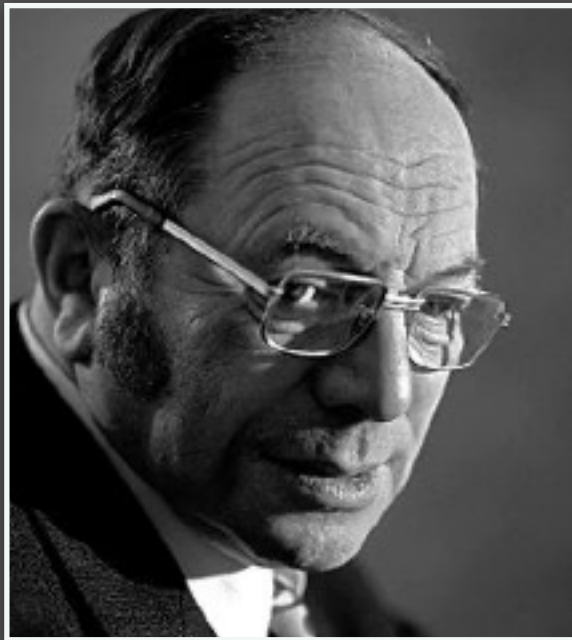


objective differentiable wrt π

$OT_c^\varepsilon(\alpha, \beta)$ is differentiable wrt α, β, c .

differentiable wrt $f, g,$



Kantorovich's Problem:

(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$

Differentiability of EOT

- Regularized formulation:



Kantorovich's Problem:

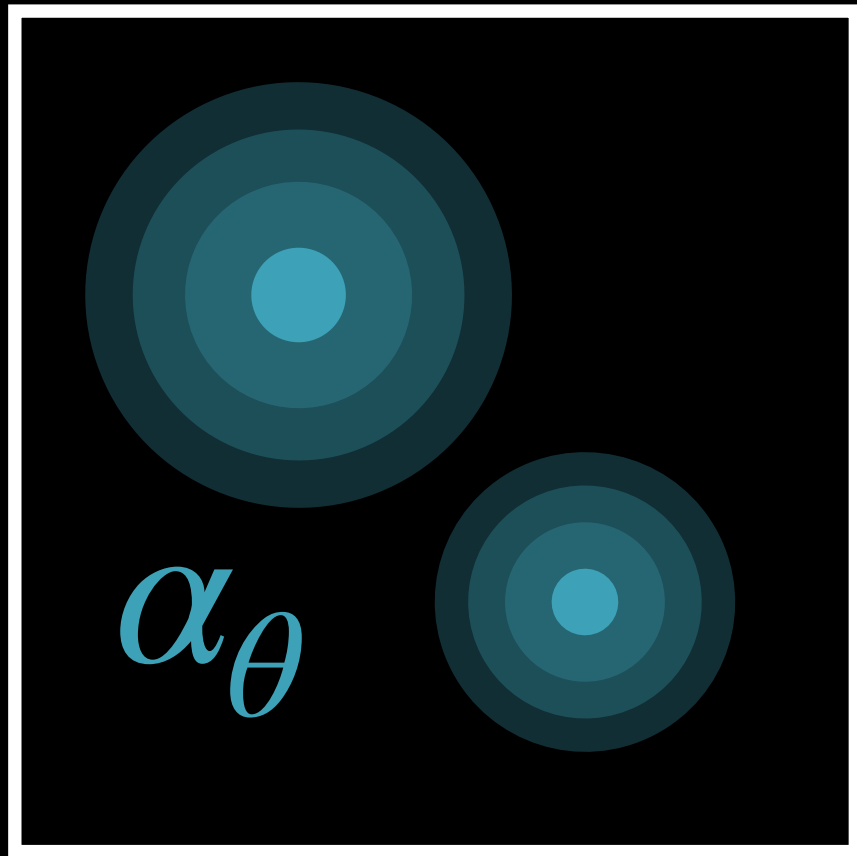
(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$

- This is strictly convex $\Rightarrow$ unique minimizer π_ε^* + **objective differentiable wrt π**
- Unique π_ε^* varies smoothly on $\alpha, \beta, c \Rightarrow \text{OT}_c^\varepsilon(\alpha, \beta)$ **is differentiable wrt α, β, c .**
- Dual problem: $\max_{f, g} \langle f, \alpha \rangle + \langle g, \beta \rangle - \varepsilon \langle e^{(f \oplus g - c)/\varepsilon} - 1, \alpha \otimes \beta \rangle$
- Dual is smooth + concave $\Rightarrow$ **differentiable wrt f, g** , optimal potentials vary smoothly

Parametrizations and Derivatives

In most interesting applications, no closed-form OT solution between α_θ, β
 $\Rightarrow$ need to **estimate OT loss from finite samples** (i.e., empirical measures)



Parametric Distribution

(our model, parametrized by θ)